ETS™

EMC TEST SYSTEMS, LR–An ESCO Company

E M C O

MODEL 6509 SERIES
Passive Loop Antenna
Installation and Operation Manual

NOTICE
vmw.atecorp.com
A MINIMUM OF $100.00 WILL BE CHARGED FOR ANY ACCESSORY NOT RETURNED WITH INSTRUMENT"""

Rentals • Sales
Test Equipment Solutions
www.atecorp.com
For Support, Please Call
Advanced Test Equipmenr
1–800–404–2832 □ ,

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P.O. Box 80589, Austin, TX 78708–0589
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http://www.emctest.com

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SHIPPING LIST

WARRANTY

T"he Electro-Mechanics Company (EMCO) warrants that our products are free from defects in materials and workmanship for a period of two years from the date of shipment. If you notify us of a defect within the warranty period, we will# at our option, either repair or replace those products which prove to be defective. If applicable/ we will also recalibrate the product.

Tliere will be no charge for warranty services performed at the location we designate. You must "however, prepay inbound shipping costs and ;any duties or taxes. We will pay outbound shipping costs for 'a carrier of our choice, exclusive of any duties or taxes. You may request warranty services to be performed at your location, but it is our option to do so. If we determine that warranty service can only be performed at your location/ you will not be charged for our travel related costs.

This warranty does not apply to:

1. **Normal wear and tear of materials**
2. **Consumable items such as fuses/ batteries, etc.**
3. **Products which have been improperly installed, maintained, or used.**
4. **Products which have been operated outside of specifications•**
5. Products which have been modified without authorization.
6. Calibration of products, *unless* necessitated by defects,

THIS WARRANTY IS EXCLUSIVE. NO OTHER WARRANTY, WRITTEN OR ORAL, IS EXPRESSED □R IMPLIED, INCLUDING BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE•

THE REMEDIES PROVIDED BY THIS WARRANTY ARE YOUR SOLE AND EXCLUSIVE REMEDIES - IN NO EVENT ARE WE LIABLE FOR ANY DAMAGES WHATSOEVER, INCLUDING BUT NOT LIMITED TO, DIRECT, INDIRECT, SPECIAL, INCIDENTAL, OR CONSEQUENTIAL DAMAGES, WHETHER BASED ON CONTRACT, TORT, OR **ANY** OTHER LEGAL THEORY.

Please contact our Sales Department for a Return Material Authorization Number before shipping equipment to us-

INSTRUCTION MANUAL
EMC□ PASSIVE LOOP ANTENNA
MODEL 6509

MODEL 6509 DESqEaPTION

Tlie EMCO Model 6509 is a 12 inch diameter Band-Switched Passive loop antenna. As a part of the EMCO Model 6505 loop antenna set it is used for Transmitting. This antenna operates in the frequency range of 1 KHz to 30 MHz.

SPECIFICATIONS

Frequency range	1 KHz - 30 MHz
Input Impedance	Nominal 50 ohms
Maximum Input Power	1 Kw
Bandswitch	Band 1 .001 - .06 MHz
	Band 2 .06 - .4 MHz
	Band 3 .4 - 1-() MHz
	Band 4 1.0 - 30.0 MHz
Loop Size	12 inches (30.4 cm)
Weight	3 lbs (1.35 Kg)
Overall height	18.75 inches (47.8 cm)
Antenna Factor	see attached

OPERATING XNSTKDCT'IONS

Select the frequency band desired by rotating the band switch on the 6509.

CAUTION: WHEN USING THE MODEL 6509 ANTENNA AS A TRANSMITTING DEVICE, BE SURE TO REDUCE AMPLIFIER POWER TO ZERO WATTS BEFORE BAND SWITCHING.

TYPICAL ANTENNA FACTORS
FOR
EMCO PASSIVE LOOP ANTENNA
MODEL 6509

FREQUENCY (MHz)	MAGNETIC ANTENNA FACTOR (db)	ELECTRIC ANTENNA FACTOR (db)
BAND 1		
.001	57.6	19.1
.002	44.8	6.3
.003	40.0	1.5
.005	36.0	7.5
.007	32.9	4.4
.010	29.8	1.4
.020	24.7	6.2
.030	22.3	3.8
.040	21.2	2.7
.050	20.5	2.0
.060	20.0	1.5
BAND 2		
.060	24.1	5.7
.070	23.3	4.8
.080	22.6	4.1
.100	21.5	3.1
.150	20.4	1.9
.200	19.8	1.3
.250	19.4	0.9
.300	19.2	0.7
.350	19.1	0.6
.400	19.0	0.5
BAND 3		
.400	16.8	8.3
.500	16.1	7.7
.600	15.3	6.9
.700	15.4	6.9
.800	15.0	6.6
.900	14.7	6.3
1.000	14.5	6.1
BAND 4		
1.000	5.1	6.6
2.000	1.3	1.2
3,000	-3.5	8.0
5.000	-7.1	4.4
10,000	-9.9	1.6
15.000	-11.6	9.9
20.000	-10.9	0.6
25.000	-11.6	9.9
30.000	-11.6	9.9

Antenna Factor Data

A.H. Systems Inc.
9710 Cozycroft Ave. Chatsworth, CA 91311
Phone (818) 998-0223 Fax (818) 998-6892
E-mail: sales@AHSystems.com
Web site: http://www.AHSystems.com

Contents List

Quantity	Part Number	Model Number	Description	SN.
1	N/A	6509	EMCO Loop Antenna 1 KHz - 30 MHz	1503
1	N/A	N/A	Calibration Manual	

A.H. Systems, Inc.
9710 Cozycroft Ave.
Chatsworth, CA 91311

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Fax: (818) 998-6892 • www.AHSystems.com

Certificate Number: 1811-2516

Certificate of Calibration

The instrument identified below has been indivichjally calibrated per the following standard(s):
IEEE Std 291-1991 - Section 2.2 Field Strength Calibration Using Loop Antennas

Manufacturer: Emco
Model Number: 6509
Serial Number: 1503
Customer: Advanced Test Equipment Corporation

Remarks:

Date Calibrated: 08-Nov-18
Calibration Interval: 24 Months
Calibration Due Date: 08-Nov-20
Temperature: 68 F (20.0 C)
Humidity: 57% Relitive Humidity

This calibration is traceable to the National Institute of Standards and Technology through documents on file in the Chatsworth Metrology Laboratory. Copies of NIST Traceabiiity and test reports are available upon request. All calibrations conform to ANSI/NCSL Z540-1-1994 and ISO/IEC 17025. Uncertainties listed are derived from methods described by NIST Tech Note 1297.

Measurement Standards and Equipment Used:

ManufactL-Irer	Mode! NL-Imber	SN	Due Date
Agilent	E5061B	MY49406483	21-Mar-19

Condition of Instrumentation

Upon Receipt: Good Condition
After Service: Within Manufacturer's Specifications

Calibration Uncertainty

95% confidence Level using a coverage factor of k=2
Correction Factor +/- 0.87 dB

JI Tim W< Jods
RF ? ichnician

Travis P. Samuels
Quality Control Manager

77?e information shown on this certificate applies only to the instrument identified above and may not be reproduced, except in full, without the wntten approval of A.H. Systems, inc. There is no implied warranty that the instrument i/w// maintain its specified tolerances during the calibration interval due to possible drift, environment, or other factors beyond our control.

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Calibration, Passive Loop Antenna

Model Number: 6509 SN: 1503 Date: 08-Nov-18

Frequency	Magnetic Correction Factor (dB)	Electric Correction Factor (dB)
1 KHz	53.033	104.533
2 KHz	46.757	98.257
3 KHz	41.127	92.627
4 KHz	38.387	89.887
5 KHz	36.163	87.663
6 KHz	34.252	85.752
7 KHz	33.022	84.522
8 KHz	31.774	83.274
9 KHz	30.858	82.358
10 KHz	29.878	81.378
12 KHz	28.469	79.969
14 KHz	27.420	78.920
16 KHz	26.533	78.033
18 KHz	25.793	77.293
20 KHz	25.237	76.737
22 KHz	24.686	76.186
24 KHz	24.176	75.676
26 KHz	23.838	75.338
28 KHz	23.455	74.955
30 KHz	23.223	74.723
32 KHz	23.025	74.525
34 KHz	22.839	74.339
36 KHz	22.595	74.095
38 KHz	22.395	73.895
40 KHz	22.223	73.723
42 KHz	22.127	73.627
44 KHz	22.044	73.544
46 KHz	21.939	73.439
48 KHz	21.816	73.316
50 KHz	21.739	73.239
52 KHz	21.637	73.137
54 KHz	21.552	73.052
56 KHz	21.559	73.059
58 KHz	21.505	73.005
60 KHz	21.433	72.933
Switch 2		
60 KHz	24.604	76.104
65 KHz	24.183	75.683
70 KHz	23.857	75.357
75 KHz	23.437	74.937
80 KHz	23.212	74.712
85 KHz	22.932	74.432
90 KHz	22.769	74.269
95 KHz	22.550	74.050
100 KHz	22.348	73.848
110 KHz	21.970	73.470
120 KHz	21.817	73.317
130 KHz	21.514	73.014
140 KHz	21.317	72.817
150 KHz	21.195	72.695
160 KHz	21.157	72.657

Frequency	Magnetic Correction Factor (dB)	Electric Correction Factor (dB)
170 KHz	21.042	72.542
180 KHz	20.889	72.389
190 KHz	20.771	72.271
200 KHz	20.664	72.164
210 KHz	20.607	72.107
220 KHz	20.556	72.056
230 KHz	20.510	72.010
240 KHz	20.475	71.975
250 KHz	20.408	71.908
260 KHz	20.353	71.853
270 KHz	20.306	71.806
280 KHz	20.257	71.757
290 KHz	20.186	71.686
300 KHz	20.165	71.665
310 KHz	20.169	71.669
320 KHz	20.196	71.696
330 KHz	20.173	71.673
340 KHz	20.139	71.639
350 KHz	20.101	71.601
360 KHz	20.057	71.557
370 KHz	19.986	71.486
380 KHz	19.930	71.430
390 KHz	19.950	71.450
400 KHz	19.967	71.467
Switch 3		
400 KHz	18.539	70.039
420 KHz	18.349	69.849
440 KHz	18.352	69.852
460 KHz	18.055	69.555
480 KHz	17.853	69.353
500 KHz	17.684	69.184
520 KHz	17.572	69.072
540 KHz	17.663	69.163
560 KHz	17.660	69.160
580 KHz	17.596	69.096
600 KHz	17.479	68.979
620 KHz	17.229	68.729
640 KHz	17.068	68.568
660 KHz	16.972	68.472
680 KHz	16.812	68.312
700 KHz	16.647	68.147
720 KHz	16.571	68.071
740 KHz	16.495	67.995
760 KHz	16.327	67.827
780 KHz	16.173	67.673
800 KHz'	16.165	67.665
820 KHz	16.125	67.625
840 KHz	15.899	67.399
860 KHz	15.702	67.202
880 KHz	15.625	67.125
900 KHz	15.548	67.048

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Calibration, Passive Loop Antenna

Model Number: 6509 SN: 1503 Date: 08-Nov-18

Frequency	Magnetic Correction Factor (dB)	Electric Correction Factor (dB)
920 KHz	15.382	66.882
940 KHz	15.209	56.709
960 KHz	15.064	66.564
980 KHz	14.988	66.488
1 MHz	14.912	66.412
Switch 4		
1 MHz	5.246	56.746
2 MHz	-0.522	50.978
3 MHz	-4.281	47.219
4 MHz	-6.742	44.758
5 MHz	-8.590	42.910
6 MHz	-7.600	43.900
7 MHz	-7.584	43.916
8 MHz	-8.460	43.040
9 MHz	-8.584	42.916
10 MHz	-9.440	42.060
11 MHz	-10.529	40.971
12 MHz	-11.771	39.729
13 MHz	-13.195	38.305
14 MHz	-14.016	37.484
15 MHz	-14.486	37.014
16 MHz	-14.236	37.264
17 MHz	-13.920	37.580
18 MHz	-13.907	37.593
19 MHz	-13.712	37.788
20 MHz	-12.993	38.507
21 MHz	-12.152	39.348
22 MHz	-11.427	40.073
23 MHz	-8.578	42.922
24 MHz	-6.023	45.477
25 MHz	•4.187	47.313
26 MHz	-3.335	48.165
27 MHz	-4.720	46.780
28 MHz	-6.164	45.336
29 MHz	-7.207	44.293
30 MHz	-7.510	43.990

Frequency	Magnetic Correction Factor (dB)	Electric Correction Factor (dB)